

Do Smart Money Concepts actually beat the market — or just the market itself?

A full translation of two LuxAlgo Pine Script indicators (Smart Money Concepts, ICT Concepts) into Python, run as event detectors across 501 S&P 500 stocks and backtested at 8 holding horizons, with standard errors clustered by ticker. **The headline result: none of it holds up** — not the aggregate claim, and not one of the 42 individual concepts either, once tested fairly.

An earlier version of this page linked a live Streamlit dashboard on a temporary Cloudflare tunnel; that link is no longer live (tunnels of that kind don't persist between sessions). Run `streamlit run streamlit_app.py` locally from the repository for the same interactive explorer, queryable against every ticker, concept, and combination in this study.

42 / 42 concepts beat a <i>zero</i>-return null — the weak test, inflated by 16 years of market drift	0 / 42 concepts beat a <i>random-entry baseline</i> at the same frequency, cluster-robust — the test that actually matters	0 / 12 sectors show positive excess return over that baseline at all
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01 Do SMC/ICT concepts outperform random entries and standard technical strategies?

No — cleanly no, and the gap between the two tests above is the reason the aggregate result looked different at first.

Every one of the 42 translated signal families — Break of Structure, Order Blocks, Fair Value Gaps, Change of Character, Liquidity Sweeps, and the rest, across both the ICT and SMC engines — produced a mean forward return significantly different from zero. On its own, that number is close to meaningless: 2010–2026 was a long bull market, and almost any signal biased toward long entries will show a positive average return purely from broad drift, independent of whether the signal contains real information.

So every ranking in this study is re-run against a second, harder bar: a random-entry strategy, backtested through *identical* mechanics, at the same trading frequency, on the same stocks, over the same window, with standard errors **clustered by ticker** rather than treated as independent per-trade draws (trades on the same stock share overlapping holding windows and are not independent observations — see Q8). Against that bar, **35 of 42 concepts differ significantly from the baseline after correcting for testing 42 hypotheses at once** (Benjamini–Hochberg FDR, $\alpha = 0.05$) — but every one of those 35 differs in the *negative* direction. Zero concepts beat the baseline.

Slice it by sector or market regime (Q5) and the picture doesn't soften: 0 of 12 sectors, and 0 of 9 regime cells, show the signal population beating the baseline with statistical significance — and unlike the earlier cut of this study, not even the point estimates are mostly positive: 11 of 12 sectors and 8 of 9 regime cells have a *negative* excess return outright. **The unconditional claim "SMC/ICT signals beat chance" is not supported by this dataset, and neither is the narrower claim that a specific minority of concepts do.** Where the data comes closest to a positive result (Q2), it stops well short of significance.

02 Which concepts provide the strongest statistical edge?

None clear the bar. Ten-day forward return, ranked by excess return over baseline — the closest anything gets.

SIGNAL	TRADES	WIN RATE	AVG RETURN	EXCESS VS. BASELINE	P (BASELINE)
smc_swing_ob_bullish_mitigated	3,421	60.1%	+1.07%	+0.23%	0.26
ict_ob_bullish_mitigated	20,705	58.0%	+0.93%	+0.10%	0.26
smc_internal_ob_bullish_mitigated	39,894	57.1%	+0.79%	-0.04%	0.03
ict_nwog_formed	421,965	57.0%	+0.76%	-0.08%	0.03
smc_equal_highs	10,547	57.3%	+0.74%	-0.10%	0.03

The two order-block *mitigation* variants — price returning to tap a block after it forms — come closest, with small positive excess over baseline. Neither reaches significance: `smc_swing_ob_bullish_mitigated` fires only 3,421 times across the whole 16-year universe, and `ict_ob_bullish_mitigated`'s p-value (0.26, already FDR-adjusted) is nowhere near the 0.05 bar. Everything else in the top 8 by excess return is already negative.

One concept does reach significance in this table's neighborhood, in the wrong direction: `ict_ndog_formed` fires 1,941,685 times — by far the largest sample of any signal in the study — and that sample size is enough to make its small negative excess return (−0.13%) statistically significant ($p = 0.030$) even though the effect itself is tiny. Large samples make small effects visible; they don't make them large.

03 Which concept combinations are most robust?

268 same-direction concept pairs co-occur on the same stock and day at least 30 times; the 60 most frequent were tested. **0 of 60** beat the baseline after correction, and confluence doesn't rescue the picture: **56 of 60** are significantly *worse* than baseline — a slightly higher failure rate than single concepts (35 of 42), not the "confluence adds information" story a naive reading of the raw win rates would suggest.

COMBINATION	TRADES	WIN RATE	EXCESS VS. BASELINE	P (BASELINE, ADJ.)
ict_ob_bullish_mitigated + smc_internal_ob_bullish_mitigated	12,384	57.6%	-0.00%	0.972
ict_ndog_formed + ict_nwog_formed	421,965	57.0%	-0.08%	0.182
ict_liquidity_buyside_pool_formed + ict_liquidity_buyside_swept	10,282	57.9%	-0.08%	0.322
ict_ndog_formed + smc_equal_highs	10,547	57.3%	-0.10%	0.226

These are the four combinations *closest* to beating the baseline, ranked by excess return — not the strongest performers, since none is positive. The best of the 60, an order-block-mitigation pairing across both engines, has an excess return of essentially zero (-0.00004, rounds to -0.00%) and a p-value of 0.97: statistically indistinguishable from doing nothing, not from beating it. One pairing worth noting for a different reason: `ict_ndog_formed + ict_nwog_formed`, at 421,965 trades, is identical to six decimal places to `ict_nwog_formed` tested alone — `nwog` (421,965 total occurrences) apparently never fires without `ndog` (1,941,685 total occurrences) also firing on the same bar, so this "combination" is a duplicate of a single concept, not genuine confluence, and inflates the 60-combination test count by at least one degenerate pair.

04 Most responsive stocks

Ranked by excess return over *that same ticker's own* random-entry baseline — not raw Sharpe, which would just reward stocks that rallied hard regardless of signal quality.

TICKER	EXCESS VS. OWN BASELINE
GL	+2.25%
DVN	+2.25%
GE	+2.12%
XYZ	+1.91%
SMCI	+1.80%
FISV	+1.68%

Least responsive — the signal population underperforms the stock's own baseline by the widest margin:

TICKER	EXCESS VS. OWN BASELINE
NFLX	-4.50%
MGM	-3.92%
NOW	-3.85%
DDOG	-3.82%
ANET	-3.68%
PLTR	-3.61%

Per-ticker baseline samples are small (~50 random entries per stock), so none of these individual rankings reach statistical significance on their own, in either direction — read this as a ranking, not a list of proven single-stock edges. Full p-values are in the data export.

05 Which sectors and regimes are most responsive?

Excess average return over the sector-matched random-entry baseline. Every sector is negative; none is statistically significant in the positive direction.

SECTOR	SIGNAL AVG.	BASELINE AVG.	EXCESS
Consumer Staples	0.21%	0.42%	 -0.20%
Utilities	0.21%	0.50%	 -0.29%
Financials	0.31%	0.64%	 -0.33%
Materials	0.28%	0.64%	 -0.37%
Real Estate	0.21%	0.63%	 -0.43%
Industrials	0.37%	0.81%	 -0.45%
Energy	0.33%	0.83%	 -0.50%
Consumer Discretionary	0.40%	1.00%	 -0.60%
Communication Services	0.28%	0.91%	 -0.64%
Health Care	0.30%	1.05%	 -0.75%
Information Technology	0.53%	1.52%	 -0.99%

Energy and Materials were the two sectors that looked positive in an earlier cut of this study — that result did not survive fixing a bug in how the per-ticker random baseline was seeded (Q8), which had been drawing near-identical "random" dates for most of the universe. With that fixed, every sector's excess return is negative. Information Technology is still the sharpest illustration of why raw returns mislead: it's the largest sector here (60 tickers), its signal population's *raw* average return is the highest of any sector (+0.53%) — and it is still the single worst result on an excess-return basis, because random long entries in IT stocks over 2010–2026 did even better (+1.52%). A high raw return is not evidence of a good signal if the do-nothing alternative did better.

BY MARKET REGIME

Regime is classified per stock, from its own trailing 200-day trend and realized volatility — not an external index.

	HIGH VOL	NORMAL VOL	LOW VOL

BULL	-0.76%	-0.38%	-0.17%
BEAR	-1.46%	-0.56%	+0.09%
SIDEWAYS	-1.32%	-0.41%	-0.86%

Cell value = signal average return minus baseline average return, at the 10-day horizon. Only one of nine cells is positive (bear market, low volatility) — and it is not statistically significant ($p = 0.90$), on a cell that holds just 15,569 of the 4.3M trades in this slice (0.4%).

06 Which concepts fail consistently?

Every bearish-direction signal in the study — `ict_bos_bearish`, `ict_fvg_bearish_filled`, `smc_fvg_bearish_formed`, and the rest — shows a *negative* average forward return, significantly so. Read carefully: because these are short-biased signals, a negative return means the position lost money, i.e. **price kept rising after the bearish signal fired**. Across a 16-year bull market, that is a coherent, honest finding rather than a bug: these signals were systematically fighting the prevailing trend.

SIGNAL	TRADES	WIN RATE	SHARPE
<code>ict_fvg_bearish_filled</code>	86,287	39.7%	-1.113
<code>ict_bos_bearish</code>	42,780	40.2%	-0.813
<code>smc_fvg_bearish_formed</code>	68,062	42.1%	-0.708
<code>smc_internal_bos_bearish</code>	16,686	41.2%	-0.698
<code>ict_volume_imbalance_bearish</code>	85,075	41.7%	-0.696

Long-biased signals fail differently, and it's worth being precise about how. `smc_fvg_bullish_formed` (+0.65% raw), `ict_ndog_formed` (+0.71% raw), and `ict_mss_bullish` (+0.67% raw) all have positive raw returns *and* are significantly worse than the baseline — the raw number and the honest comparison point in opposite directions. `ict_nwog_formed` is the one exception among long-biased signals that's genuinely indistinguishable from baseline rather than significantly worse ($p = 0.20$). A positive raw return with no baseline comparison is not evidence of an edge — it can just as easily describe a signal actively working against you at a smaller loss than the market's average gain.

07 How stable is this across time and conditions?

No. The regime grid in Q5 has one positive cell out of nine — bear markets during low-volatility stretches — and it is both statistically insignificant ($p = 0.90$) and a narrow slice of the data (15,569 of 4.3M trades in this cut, 0.4%). Every other combination of trend and volatility regime shows the signal population trailing a plain random entry, several by more than a full percentage point. This is the sharper version of the "0 of 42 concepts beat baseline" headline, not a separate caveat on it: there is no regime, sector, or time slice in this dataset where the aggregate signal population comes out ahead with statistical support.

08 Method, in brief

- **Translation.** Both Pine scripts were read line by line and every concept ported to Python by hand — pivot detection, order block formation and mitigation, fair value gaps, liquidity pools, structure breaks — with overlapping vocabulary between the two engines (three different working definitions of "BOS," for instance) kept as separate signal families rather than merged.
- **Detection.** Every detector ran against 501 of 503 S&P 500 constituents, daily bars, 2010–2026: 4,326,101 discrete events across 42 signal families.
- **Backtest.** Every event was carried forward 1, 2, 3, 5, 10, 20, 40, and 60 trading days — 34,452,258 trades total. Entry uses only the signal bar's own close; every exit price and adverse/favorable excursion is computed strictly on bars after entry. No look-ahead by construction.
- **The baseline correction.** A first pass reported all 42 concepts as "significant" — using only a zero-return null. That was a red flag, not a result: it meant a benchmark comparison was missing, not that the signals worked. A random-entry baseline was built and backtested through the identical pipeline, and every ranking — concepts, combinations, individual stocks, sectors, regimes — was re-tested against it.
- **Two further corrections, applied after the first published cut of this study.** (1) The per-ticker random baseline was seeded from a hash of each ticker's first cached bar date; because most tickers share that date, roughly 84% of the universe was drawing *the same* "random" dates rather than independent ones, and the fix (a stable per-ticker seed) changed the baseline meaningfully — it's the reason the sector table in Q5 no longer has any positive entries. (2) Every significance test now clusters standard errors by ticker rather than treating each trade as an independent draw: trades on the same stock share overlapping holding-period windows, and a naive per-trade test understates the true standard error as sample size grows. Sector and market-regime rankings did not carry a significance test at all in the first published cut — a raw excess-return number with no test of whether it differs from noise — and now do.
- **Also fixed:** the SMC order-block detector had its bullish/bearish extreme-candle selection swapped relative to the source script (picking the highest high for bullish blocks and the lowest low for bearish ones, the reverse of the source's demand/supply logic), and both engines' order-block detectors included the current breakout bar in their search window where the source excludes it. Both are corrected; the numbers on this page reflect the corrected detectors.
- **Significance.** Cluster-robust confidence intervals, cluster-robust two-sample significance tests against the baseline, and Benjamini–Hochberg false-discovery-rate correction throughout, at $\alpha = 0.05$. Clustering corrects for within-ticker correlation; it is not two-way (ticker-by-date) clustered, so cross-ticker correlation on shared market-wide dates is not separately corrected for — a conservative gap, since implementing it would tighten these results further, not loosen them.

Two concepts referenced in general ICT literature — Rejection Blocks and Optimal Trade Entry — have no corresponding logic in either source script and were not implemented. No transaction costs or slippage are modeled; these are in-sample results across the full study window, not a walk-forward or held-out test. The universe reflects current-day S&P 500 membership applied retroactively across 2010–2026, not a point-in-time-correct historical constituent list — constituents removed from the index during the window are absent, and recent additions carry their full available trading history despite not having been index members for most of it. A companion academic working paper with full literature review and methodology writeup is at [docs/ssrn_paper.pdf](#) in the repository. Source indicators © LuxAlgo, CC BY-NC-SA 4.0. This is a research study, not investment advice.